

DNA Evolution Feed Locus at the FAI Hand-Off Boundary

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Paper 3’s Claim 4 specifies a four-locus evolution-feed mechanism at the Full Aspect Integration (FAI) hand-off boundary. This note formalizes the second locus: the DNA evolution feed locus. Where home governance authorizes it, DNA-layer content from another Self’s contributed aspects can be selectively absorbed into the receiving Self’s home DNA. Absorption operates exclusively through Paper 2’s directed selection mechanism — human-authorized, versioned, and provenance-attributed. The locus operates entirely under home governance authority: the contributing Self’s governance cannot compel absorption, and the joint authority that governs the shared substrate does not extend to changes in the home substrate. Selectivity — governance’s power to absorb some candidate DNA content and decline other — is the architectural feature that distinguishes this locus from an automatic transfer mechanism and from a non-selective ingestion mechanism. This note states the sub-commitment precisely, covers the three-step absorption process, establishes the home-governance sovereignty boundary, identifies the four failure modes the commitment defends against, and provides an operational test.

1. The sub-commitment

D1.18 states: where home governance explicitly authorizes it, DNA-layer content from another Self’s contributed aspects within the shared substrate can be selectively absorbed into the receiving Self’s home DNA through Paper 2’s directed selection mechanism, operating entirely under home governance authority, with full provenance of the FAI event carried in the resulting DNA version

history entry.

Three properties are jointly constitutive of this sub-commitment, and each must hold simultaneously for the commitment to be satisfied:

Selective. Home governance identifies which candidate DNA content from the shared substrate to absorb and which to decline. Selective absorption means that governance's decision is per-candidate, not binary over the whole available set. The authority to decline is as architecturally significant as the authority to absorb.

Through directed selection. Absorption is not a separate mechanism introduced at the FAI boundary. It is Paper 2's directed selection event receiving a new class of input. Every governance property of directed selection — human authorization, DNA version history entry, provenance attribution, the exercise of the modify right — holds for DNA content absorbed from an FAI event, exactly as it holds for DNA content authored by internal governance analysis or generated by action-feedback proposals.

Entirely under home governance authority. The contributing Self's governance has no authority over whether the receiving Self absorbs any DNA content. The joint governance authority established at FAI inception covers the shared substrate and its contents; it does not extend into the receiving Self's home substrate. The boundary between shared-substrate authority and home-substrate authority is the governance sovereignty line D1.18 formalizes.

2. Where this locus sits

Paper 3's Claim 4 establishes four loci at the FAI hand-off boundary where FAI outputs enter participating Selves' existing Paper 2 evolution machinery. D1.17 formalizes the action-feedback locus — FAI outcomes becoming action-layer content that home action-feedback evolution processes. D1.18 formalizes the second locus: the DNA evolution feed locus. The remaining two loci are addressed by subsequent D1 notes.

At dissolution, the shared substrate carries two kinds of content: action-layer records of what was done during the FAI event, and DNA-layer content from contributing Selves' aspects. The action-feedback locus handles the first kind. The DNA evolution locus handles the second kind, but only where governance authorizes it and only through the selective absorption process described here.

The instinct evolution locus is absent by design. Instinct-layer content — LLM weights and instinct-layer substrate content — does not cross the inter-Self perimeter. This is inherited from Paper 2's instinct/reasoning separation, extended through Paper 3's exchange-bounding commitment. The DNA evolution feed locus therefore operates within a bounded domain: DNA-layer content only, home-governance-authorized absorption only.

3. The three-step absorption process

Step 1 — Candidate identification. At or following FAI dissolution, the receiving Self's governance reviews DNA-layer content from the shared substrate that is candidate for absorption. The candidate set is determined by the sharing-scope configuration that governed the FAI event: governance specifies, at configuration time, which aspects each Self contributes and with what provenance carry-over depth. The candidate set for absorption is therefore itself a product of prior governance decisions, not an arbitrary slice of the contributing Self's DNA.

Governance identifies the candidate set by inspecting shared-substrate content at the FAI hand-off boundary. This is an exercise of the inspect right over the shared substrate, which home governance holds under the joint authority that governed the FAI event. Identification does not commit to absorption; it produces a reviewed candidate set against which governance then exercises selection authority.

Step 2 — Governance selection. Home governance reviews the candidate set and decides, per candidate, what to absorb and what to decline. This decision is the exercise of the modify right (Paper 1, Claim 3) at the FAI hand-off boundary: governance decides what enters the home substrate's DNA layer.

Selection authority is unrestricted within home governance. There is no architectural obligation to absorb any candidate, and there is no architectural default that treats non-response as approval. If governance makes no selection decision, no DNA content from the FAI event enters the home substrate. The architecture's default is no absorption; absorption requires an affirmative governance decision.

Selection decisions may be made by human review of each candidate, by governance-authored rules that pre-configure absorption criteria for a class of DNA content, or by a combination of both. What must hold in all cases is that the resulting absorption is traceable to human governance authority: either a human decided directly, or the rules that determined the decision were human-authored and remain subject to the three rights (inspect, modify, override).

Step 3 — Directed selection event. Each approved absorption is implemented as a directed selection event in the receiving Self's home governance. A directed selection event in Paper 2's architecture has four properties that carry through unchanged at the FAI DNA evolution locus:

Human authorization. The event is authorized under home governance authority. Whether a human acts directly or acts through governance-authored rules, the authorization chain terminates at human authority.

DNA version history entry. The directed selection event produces a versioned entry in the receiving Self's DNA lineage. The absorbed content becomes part of the home substrate's versioned history at the moment of the event, exactly as internally-authored DNA changes do.

Provenance of FAI origin. The version history entry carries provenance that identifies the FAI event as the source of the change and, within that, the contributing Self's aspect from which the candidate DNA content came. This provenance is not optional; it is what makes path retraceability (Paper 1) hold at the DNA evolution feed locus.

Subject to all Paper 2 directed selection governance requirements. No governance requirement that applies to directed selection in the intra-Self context is suspended at the FAI DNA evolution feed locus. The locus adds a new class of input to directed selection; it does not modify the governance shape of directed selection.

4. Home governance sovereignty

The DNA evolution feed locus is the architectural commitment that defines the governance sovereignty boundary for FAI participation. Two authority structures operate in proximity at the FAI hand-off boundary, and the commitment specifies precisely where each ends.

Joint authority governs the shared substrate. During an FAI event, the participating Selves operate under a joint authority structure that governs the shared substrate: what aspects are contributed, how the shared substrate is organized, what conflict-handling tiers apply, what remains after dissolution, and what provenance carry-over depth applies. This joint authority is the governance of the FAI event itself.

Home governance governs the home substrate. Joint authority does not extend into the home substrate. The decision of whether to absorb DNA content from the FAI event, which content to absorb, and what directed selection event implements that absorption — all of these decisions are made exclusively by home governance. The contributing Self's governance authority terminates at the shared-substrate boundary. The joint authority of the FAI event terminates at the same boundary when the question is what changes enter the home substrate.

This boundary is what makes FAI participation architecturally safe for organizations that have governance sovereignty commitments. Participation in an FAI event does not, by the architecture's design, require accepting any changes to the home DNA. An organization can engage in full FAI participation — contributing aspects, operating within the shared substrate, using FAI outputs — without the architecture compelling any DNA changes at home. Whether to evolve the home DNA based on what an FAI event revealed is a governance decision that remains entirely internal.

The contributing Self has no recourse within the architecture if the receiving Self declines absorption of DNA content the contributing Self considers valuable. This asymmetry is intentional: governance sovereignty requires that the asymmetry run in this direction, not the reverse.

5. Inheritance from Paper 2 directed selection

The DNA evolution feed locus is Paper 2's directed selection mechanism receiving a new class of input. The mechanism is unchanged; the input source is new.

In the intra-Self context (Paper 2), directed selection receives candidates from two sources: internal governance analysis that identifies improvements to the DNA, and action-feedback evolution proposals that surface operational evidence as candidate DNA changes. In both cases, governance reviews, selects, and implements through the directed selection event.

At the FAI DNA evolution feed locus, a third candidate source is added: DNA-layer content from another Self's contributed aspects within the shared substrate. Governance reviews this candidate set through the same selection process, implements approved absorptions through the same directed selection event, and produces the same version history entry with the same governance properties. The only new element is the provenance carried in the version history entry, which must name the FAI event and the contributing aspect as the source.

This inheritance structure is what the "INHERITANCE FROM PAPER 2" framing names. It is not a claim that FAI adds nothing to the evolution architecture. It is a claim that what FAI adds at the DNA evolution locus is a new candidate source for an existing mechanism, not a new mechanism. The governance shape — human authorization, versioning, provenance, the modify right — is the same governance shape Paper 2 established for directed selection in the intra-Self context.

This inheritance is also what makes the DNA evolution feed locus architecturally coherent with the rest of the trilogy. The CKS pattern's commitment to governed substrate evolution does not require a separate governance regime for each new context in which evolution occurs. It requires that evolution, wherever it occurs, go through governed mechanisms with the same governance properties. The FAI DNA evolution locus extends that requirement to the inter-Self boundary without introducing a new governance regime.

6. Four failure modes the sub-commitment defends against

D1.18 formalizes an architectural commitment, not merely a desirable property. The commitment has traction because four specific failure modes exist in architectures that do not make it, and each failure mode undermines a distinct governance property.

Failure mode 1 — Automatic DNA absorption. An architecture in which FAI participation automatically updates home DNA without an explicit governance authorization step does not satisfy D1.18. Automatic absorption eliminates Step 2 of the absorption process: there is no governance selection decision, only an automatic transfer. This breaks the human governance commitment at the DNA evolution boundary and makes the home substrate's DNA layer subject to change by FAI participation alone.

Failure mode 2 — Non-selective absorption. An architecture in which all contributed DNA content from an FAI event enters the home DNA — without governance having the ability to accept

some and decline other — does not satisfy D1.18. Non-selective absorption treats the DNA evolution feed as a transfer protocol rather than a governed improvement process. It removes governance’s per-candidate selection authority, reducing governance to a binary gate (participate or not participate) rather than a selection authority over candidate content.

Failure mode 3 — Non-home-governed absorption. An architecture in which joint authority or the contributing Self’s governance authority determines what DNA content enters the receiving Self’s home substrate does not satisfy D1.18. Non-home-governed absorption places authority over home substrate changes outside home governance, which is the structural equivalent of making FAI participation a condition for accepting DNA changes from other Selves. This failure mode is the most direct violation of the sovereignty-preserving property.

Failure mode 4 — Unattributed DNA changes. An architecture in which DNA content absorbed from an FAI event does not carry provenance of its FAI origin in the receiving Self’s DNA version history does not satisfy D1.18. Unattributed absorption breaks path retraceability (Paper 1) at the DNA evolution feed locus. An observer of the receiving Self’s DNA history would encounter DNA changes whose origin within the home governance record is not distinguishable from internally-authored changes — a breakdown of the traceability commitment the trilogy relies on throughout.

Each failure mode is independently sufficient to disqualify an architecture from satisfying D1.18. An architecture that avoids three of the four still fails the commitment.

7. Operational test

For a DNA change in a participating Self’s home substrate that originated from a FAI event, the following three questions define the operational test for D1.18:

Question (a) — Home governance authorization. Does the directed selection event record show that home governance authorized the DNA change? The record must be traceable to a human governance authorization — either a direct human decision or a governance-authored rule that was itself human-authorized and remains subject to the three governance rights (inspect, modify, override). Authorization by the contributing Self’s governance alone, or by the joint authority of the FAI event alone, does not satisfy this question.

Question (b) — Provenance linking to FAI event and contributing aspect. Does the DNA version history entry carry provenance that identifies (i) the FAI event as the source of the change and (ii) the contributing Self’s aspect from which the candidate DNA content came? Both provenance elements must be present. Provenance that identifies only the FAI event without identifying the contributing aspect is incomplete; provenance that identifies neither is absent.

Question (c) — Consistency with Paper 2 directed selection governance. Is the DNA version history entry consistent with Paper 2’s directed selection governance? This means: the entry is

versioned in the receiving Self's DNA lineage (not a side-channel record); the entry is the product of a selection decision (not an automatic transfer); and the entry is subject to all governance requirements that apply to directed selection in the intra-Self context.

An affirmative answer to all three questions means the DNA change at issue satisfies D1.18. A negative answer to any one question means the DNA evolution feed locus did not operate correctly for that change — and the architecture's governance sovereignty commitment was not preserved at that hand-off boundary.

8. Conclusion

The DNA evolution feed locus formalizes the architecture's answer to a specific question: when an FAI event generates DNA-layer content that a participating Self could learn from, how does that content enter the receiving Self's home DNA in a way that preserves governance sovereignty?

The answer has three parts. First, it enters only if home governance authorizes it — not automatically, not by joint authority, and not at the contributing Self's direction. Second, it enters through directed selection — the same mechanism Paper 2 established for governed DNA evolution in the intra-Self context, extended to receive a new candidate source at the FAI hand-off boundary. Third, it enters with provenance — the receiving Self's DNA version history records the FAI origin of every absorbed change.

Together, these three parts make FAI participation a governed improvement pathway for DNA evolution, not a compelled update mechanism. The receiving Self can participate fully in FAI events, observe and retain access to another Self's DNA-layer contributions in the shared substrate, and still decide — through its own governance authority, per candidate, under its own version history discipline — what if anything from that event enters the home DNA.

This is the governance sovereignty property that makes FAI architecturally safe at the DNA evolution boundary. It closes the gap between inter-Self coordination (which requires sharing) and governance sovereignty (which requires that the home substrate's evolution remain under home authority).

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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